Sravanthi Desham

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Profile

Network Engineer with 4+ years of hands-on experience developing and testing control-plane features on Broadcom-based switch platforms (Katana, Trident+, Qumran). Skilled in L2/L3 networking protocols, Broadcom bcmsdk integration, and Python-based automation using Pytest for testing switching and routing functionalities. Strong in CLI/NOS feature development, debugging, and cross-functional collaboration to deliver reliable and efficient network software solutions.

Experience

Connectivity & Network Engineer

Capgemini Engineering **June 2022 – July 2025**

- Developed, maintained, and enhanced a C-based control-plane stack and associated switching/routing (L2/L3) modules, including STP, ELPS, ERPS, LLDP, and L2CP, along with control-plane routing protocols such as OSPF and BGP, and feature components BFD and QoS, for an OTN Systems client operating carrier-grade networks built on Broadcom ASIC platforms (Qumran, Trident+, Katana), with a strong focus on MPLS-TP transport networks.
- Implemented multi-threading, synchronization primitives, shared memory, and IPC mechanisms to enable reliable inter-module communication, performed protocol enhancements, and developed custom packet-handling features based on client requirements.
- Implemented and validated port-based and VLAN-based services, enabling flexible provisioning and improved traffic segregation across customer and management domains.
- Contributed to the development of Ethernet service features including VPLS, VPWS, and ECFM, ensuring carrier-grade performance and OAM (Operations, Administration, and Maintenance) compliance.
- Integrated control-plane components with Broadcom BCMSDK, enabling efficient programming of hardware-level forwarding tables (MAC, FDB, VLAN, routing) and ensuring seamless hardware–software synchronization.
- Worked on field-processor configuration and implementation for packet classification, filtering, and forwarding on Broadcom ASICs, enhancing data-path processing and feature scalability.
- Contributed to white-box testing and development of Broadcom-based Qumran platform, validating hardware–software integration, packet forwarding, and ASIC-level functionality.
- Developed and maintained Python/Pytest automation frameworks to validate L2 protocol behavior, switching operations, and packet processing, achieving ~65 % automation coverage and significantly reducing manual test effort.
- Enhanced the Automation Test Framework (ATF) by migrating legacy setup/teardown logic to Pytest fixtures, improving maintainability, runtime efficiency, and reusability across modules.
- Utilized Jenkins to monitor and track weekly regression results and ATF module failures, ensuring consistent visibility into automation of health and test stability.
- Developed CLI functionality using SNMP MIBs, leveraging both standard and client-specific MIBs for configuration, monitoring, and control of network elements.
- Documented network configurations, protocol enhancements, and feature updates using Confluence, ensuring accurate communication and traceability of development activities.

- Performed functional and interoperability testing with Wireshark, Ixia, and Spirent, verifying label-switched paths, protocol messages, and forwarding consistency.
- Collaborated with QA, automation, and ASIC teams to align control-plane features with data-plane design and system-level testing strategies.
- Conducted detailed debugging and root-cause analysis using Broadcom bcm shell, Linux utilities, and protocol logs to isolate and resolve software and hardware issues.
- Used Git/Bitbucket for version control and JIRA for defect tracking, ensuring timely delivery, proper documentation, and quality-driven release management.

Software Engineering Intern

Capgemini **July 2021 – June 2022**

- Developed and debugged Linux-based network software in C, applying system programming concepts including processes, file I/O, and network sockets.
- Implemented a TFTP client and server over UDP to enable reliable file transfer operations.
- Applied TCP/IP and UDP protocols to build and test low-level network functionality.
- Gained hands-on experience in network programming and Linux system fundamentals.

Technical Skills

Programming Languages: C, C++, Python, Java
Database & Query Language: SQL
Networking Expertise: Routing and Switching (L2/L3) Protocols, Control-Plane Development, NOS Integration, Broadcom bcmsdk, SNMP, NETCONF, Yang
Automation & Frameworks: Pytest, Python Scripting, Network Test Automation, Regression Frameworks
Tools & Utilities: Wireshark, Ixia, Spirent, JIRA, Bitbucket, Git, Vim Editor, PuTTY, Visual Studio Code (VSCode), GDB
Core Concepts: Data Structures & Algorithms (DSA), OOPS, OS Concepts, ONL (Open Network Linux), Network Stack Debugging, CLI Development
Additional Skills: Familiar with Linux and Windows operating systems, Working ability in Agile environments

Certifications

- CCNA Routing and Switching: Introduction to Networks
- Python Fundamentals and Data Structures – University of Michigan (Coursera)

Recognitions

Received "Outstanding Contribution in Delivery" Award for contributions to ATF script enhancements that improved test suite performance and increased coverage through fixture-based implementation.

Education

Bachelor Of Technology (B.Tech.) In Electronics and Communication Engineering
Anurag Group of Institutions, Hyderabad, India | 2018 - 2022 | CGPA 9.18/10